Change of bound variable

The Open Logic Project contributors

openlogic:first-order-logic/axiomatic-deduction:local-snapshot-20261013; CC BY 4.0

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-first-order-logic-axiomatic-deduction/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: in-source stubs lem:free-for, thm:Generalization, thm:deduction-thm (all this batch); Taut is the propositional-tautology closure rule of the calculus.

Editorial note. The label lem:ChangeBdVar names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lem environment in the unit source.

Allowed standard prerequisites (editorial).

- Free-for substitution lemma. Same unit, \ollabel{lem:free-for} (source line 1359, this batch). First step: y is free for x in subst(A,c,x).
- Generalization and the deduction theorem. Same unit, thm:Generalization (this batch) and thm:deduction-thm (this batch). Middle steps: generalize over y under the not-free-in hypothesis; unbox via the deduction theorem twice.

Source reference. Original lemma lem-free-for: sources/olp-first-order-logic-axiomatic-deduction/pinned-original-source.tex, source lines 1359–1359. Prerequisite pointer invoked at proof line(s) 1381; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1 (Change of Bound Variable)**.** *If x and y are not free in A and they are both free for c in A, then $\vdash \forall x\, A[x/c] \equiv \forall y\, A[y/c]$ (and the proof does not involve c).*

Proof. From the hypotheses, y is free for c in A and x is not free in A, so by the previous lem-free-for, y is free for x in $A[x/c]$. It follows that $\forall x\, A[x/c] \to A[x/c][y/x]$ is an axiom (**Ax1**). Since x is not free in A, we have $A[x/c][y/x] = A[y/c]$, so that

$$\vdash \forall x\, A[x/c] \to A[y/c],$$

and by the Deduction Theorem $\forall x\, A[x/c] \vdash A[y/c]$. Since y is not free in A, it also not free in $\forall x\, A[x/c]$, so that by Generalization $\forall x\, A[x/c] \vdash \forall y\, A[y/c]$, and the Deduction Theorem again gives $\vdash \forall x\, A[x/c] \to \forall y\, A[y/c]$. The proof of $\vdash \forall y\, A[y/c] \to \forall x\, A[x/c]$ is perfectly symmetric, so that the conclusion follows by Taut. $\square$